Alberto Giuseppe Perotti

alberto.g.perotti@gmail.com // +39 347 740 3464 // Linkedin // Google Scholar

About me

I am a researcher in satellite communications with the ASTRACOM Research Group at the Department of Electronics and Telecommunications, Politecnico di Torino. From 2013 to 2025, I was a Principal Research Engineer at Huawei Technologies, where I conducted research on 4G and 5G wireless networks with a strong focus on 3GPP standardization, and coordinated collaborative research projects with academia. Previously, I spent nearly a decade as a researcher in academia, researching on wireless broadband terrestrial and satellite communications.

I hold a Ph.D. in Electronics and Communications [2003] ad a *Laurea* degree in Telecommunications Engineering [1999] from Politecnico di Torino, Italy.

An extended version of my CV can be found here.

Expertise

Wireless networks. PHY/MAC layer of 4G and 5G radio access networks.
3GPP Standardization. Contributed to 4G and 5G radio access networks layer 1 standardization [2013-2025]. Participated to 3GPP RAN1 (PHY layer) standardization meetings [2015-2017].
Modeling and simulation of transmission systems. Developed several link/system-level wireless system simulators for performance evaluation of lower-layer (PHY/MAC) functionalities in wireless networks.
Software Tools and Programming skills. Matlab, Python, C/C++; Keras/Tensorflow, Pytorch, nVidia GPU configuration/maintenance; GIT, SVN; Microsoft Office; Windows, Linux; assembly languages of Intel x86 processors, TI's C6000 DSPs, PIC microcontrollers, Z80.
Management and Coordination of Research Projects. Project manager of Huawei internal research projects in cooperation with Academia:

- *5G NR FR1/FR2 V2X communications*, cooperation with Politecnico di Milano, Italy;
- *Iterative error correction codes for channels with noisy feedback*, *Iterative error correction codes for realistic wireless channels*, both in cooperation with Imperial College London, UK.

Quantum Error Correction. Developing *qLDPCsim*, an open-source quantum LDPC error correction codes simulator.
Teaching. Lectured undergraduate and graduate courses at Politecnico di Torino for nearly a decade.
Editorial Experience. Associate Editor-in-Chief of IEEE Communications Magazine for [2022-2023].
Languages. *Italian*: mother tongue; *English*: professional working proficiency; *French*: basic.

Research

Error correction codes: Low Density Parity Check (LDPC) codes, polar codes, sparse superposition codes, parallel (turbo) and serial concatenations of convolutional codes, algebraic codes (Reed-Muller, Reed-Solomon, BCH, etc.), nonlinear codes, codes for identification. **AI/Machine-learning**: deep learning-based design and decoding of ultra-reliable error correction codes. **Modulations and waveforms**: constant-envelope continuous phase modulations (CPM); trellis-coded modulation; adaptive coded modulation methods; waveforms for simultaneous information and power transfer (SWIPT); coded modulations for direct satellite uplink. **Multiple access**: orthogonal/non-orthogonal multiple access; interleaver-division multiple access; multiuser detection and interference cancellation. **V2X communications.** High-gain beamforming and opportunistic relay for V2V and V2I communications. **Cognitive radio and spectrum sensing.** Sensing of DVB signals, signal processing software optimization of sensing algorithms, RF immunity evaluations. **Software-defined radios**: optimization of real-time wireless transceiver signal processing algorithms on digital signal processing platforms.
Corresponding patents: 18 granted patents, 27 pending applications (at time of writing).

Education

Aug 2003. Ph.D. degree in Electronics and Communications, Politecnico di Torino. Dissertation on **Design of concatenated convolutional codes with interleavers and their decoder implementation on multi-DSP systems**.
Supervisor: prof. S. Benedetto.

Mar 1999. Laurea[1] degree in Telecommunications Eng., Politecnico di Torino. Monograph on **Design and real-time DSP implementation of a CCSDS turbo decoder**.
Supervisors: prof. S. Benedetto, prof. A. Serra.

Employment

Jan 2026 - present. Researcher, ASTRACOM Research Group at Politecnico di Torino. Group led by prof. Roberto Garello.

Jun 2013 - Dec 2025. Huawei Technologies Sweden/Italy. Principal research engineer, research on PHY layer of 4G and 5G radio access networks; contributing to 3GPP standardization in Radio Access Networks Working Group on Radio Layer 1 - Physical layer (RAN1); management of research projects.

Apr 2011 - May 2013. CSP–ICT Innovation, Torino (Italy). Head of the *Wireless Communications and Networks* research unit.

Mar 2010- Mar 2011. Assistant Professor at Università e-Campus, Novedrate (Como), Italy. Research on spectral/energy efficient wireless communication systems.

Aug 2003–Mar 2010. Post-doctoral researcher at Politecnico di Torino, Italy. Research on PHY algorithms and architectures for wireless multimedia broadband transmission, adaptive coding and modulation system for satellite communications.

Jan-Oct 2002. Sequoia Communications (no longer in business), San Diego (CA), USA. Staff member of baseband signal processing branch in Los Angeles. Development of baseband signal processing algorithms for a prototype 3G (UMTS) receiver. Head of division/supervisor: prof. Dariush Divsalar (JPL).

Jan-Aug 2002. University of California, Los Angeles (CA), USA. Visiting scholar in the Electrical Eng. Dept. Research on design of serially concatenated convolutional codes. Supervisor: prof. Richard D. Wesel.

Teaching and student supervision

I taught several graduate and undergraduate courses at Politecnico di Torino and in other Universities, as part of Politecnico's joint programs with those institutions. Supervised/co-supervised twelve master thesis works [2009-2013].

Editorial Experience

IEEE Senior Member since 2014.
Served as editorial board member of IEEE Communications Magazine:

- *Associate Editor in Chief* from Nov 2021 to Dec 2023;
- *Lead Editor* of Mobile Communications and Networks Series from Sep 2019 to Nov 2021.

[1] In the Italian academic system before the 1999 reform, the *laurea* degree was the highest academic degree obtainable before Ph.D. It is considered equivalent to a masters' degree.

- *Associate Technical Editor* from Oct 2014 to Aug 2019;

Serving as a journal/magazine article reviewer and conference TPC member/reviewer for numerous journals and conferences.

Appointments/board membership

- Springer book series Textbooks in Telecommunication Eng.: member of Editorial Advisory Board, 2024-now.
- CCABA - Advanced Broadband Communications Center at Universitat Politécnica de Catalunya: member of External Advisory Board, 2021-now.
- French National Research Agency, project proposal reviewer, 2020.

GDPR statement

I authorise the processing of personal data contained within this CV, according to GDPR (EU) 2016/679, Article 6.1(a).